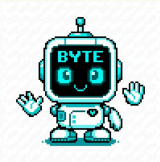
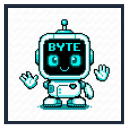
GYANBIT

USER MANUAL



BUILD IT. CODE IT. PLAY IT.

Version 1.0 | TC26-GF-005 | MMCOE Pune 2026



"Hey there! I'm BYTE — your GyanBit guide! 🤖 I'll be popping up throughout this manual to give you tips, tricks, and secret knowledge. Whenever you see me, pay attention!"

CHAPTER 1 — WELCOME TO GYANBIT

Welcome to GyanBit — India's first DIY 8-bit handheld game console built specifically for students like you.

This isn't just a toy. You are going to:

- **ASSEMBLE** a real electronic device with your own hands
- **WRITE** actual JavaScript code that runs on real hardware
- **CREATE** your own video games from scratch
- **UNDERSTAND** how every device you've ever used actually works inside

The screen you are looking at right now? Your code draws every single pixel on it. The sound that comes out of the speaker? Your code decides exactly what frequency to play and for how long. The buttons? Your code decides what happens when you press them.

That's not magic. That's embedded systems. And by the time you finish this manual, you'll understand it completely.

BYTE says: *"Fun fact — the original Game Boy from 1989 had almost the same screen resolution as GyanBit. The games were just as fun. Yours will be too!"* 🎮

CHAPTER 2 — FOR PARENTS & GUARDIANS

A LETTER FROM THE GYANBIT TEAM

Dear Parent or Guardian,

Thank you for choosing GyanBit for your child. We believe the best way to learn technology is to build it yourself — not watch someone else explain it on a screen.

What your child will genuinely learn:

GyanBit is not a toy that teaches "coding" through drag-and-drop puzzles. When your child writes a game on GyanBit, they are writing real JavaScript. Concepts encounter here directly map to CBSE curriculum for Classes 7 through 12.

Is it safe?

Yes, completely. The system runs at 3.3V logic — far below any threshold that could cause harm. There is no soldering involved. All components snap together with color-coded connectors.

Screen time and ergonomics:

We recommend sessions of 45–60 minutes with a 10-minute break. Encourage your child to sit at a proper desk while coding.

Is it a safe online environment?

GyanBit Studio does not require an account, has no social features, and no ads. The game editor is a completely self-contained tool.

CHAPTER 3 — WHAT'S IN THE BOX

When you open your GyanBit kit you should find:

#	Component	What it does
1	Raspberry Pi Pico	The brain — runs all your code
2	1.3" OLED Display	The screen — 128x64 pixels
3	PAM8403 Amplifier	Makes sound happen
4	RB-20 Speaker	Where sound comes out
5	TP4056 Module	Charges battery safely
6	Tactile Buttons x8	D-pad and A/B controls
7	PCB Carrier Board	Holds everything together
8	This Manual	You're already using it

> ***BYTE says:*** *"Handle the Pico by its edges. The gold pins on the sides are how it talks to everything else — don't bend them!"* ⚡

Missing something? Contact us at tc26gf005@gyanbit.vercel.app with your order details.

CHAPTER 4 — MEET YOUR CONSOLE

Controls:

- **D-Pad (↑↓←→):** Move characters, navigate menus
- **A Button:** Confirm, jump, shoot, interact
- **B Button:** Cancel, dash, special action
- **USB-C Port:** Power + programming
- **BOOTSEL Button:** Hold while connecting to enter flash mode

Indicator LED: The small LED on the Pico blinks during code upload and stays solid when running normally.

CHAPTER 5 — ASSEMBLY GUIDE

Time needed: 20–30 minutes | **Tools needed:** None.

"Before you start — find a clean flat surface and open this page on a second screen or phone so you can follow along while you build!" 🔧

STEP 1 — SNAP THE PICO ONTO THE BOARD

Line up the pins carefully — the USB connector on the Pico should face the USB cutout on the board. Press down firmly and evenly until it clicks into place.

STEP 2 — CONNECT THE OLED DISPLAY

The OLED module has 4 pins (GND, VCC, SCL, SDA). Match the pins to the labeled slot on the carrier board and press gently until it seats fully.

STEP 3 — CONNECT THE PAM8403 AMPLIFIER

Match the IN/OUT/5V labels on the green amplifier module to the labels on the carrier board. It snaps in firmly.

STEP 4 — CONNECT THE TP4056 MODULE

The blue charging module snaps into the power slot with the USB-C port facing outward.

STEP 5 — CONNECT THE SPEAKER

Slot the speaker into the circular recess and connect the wires to the + and − pads on the carrier board.

STEP 6 — INSTALL THE BUTTONS

Press each tactile button into its slot. They snap in with a satisfying click. There are 4 for the D-pad and 4 for the action buttons.

STEP 7 — CLOSE THE ENCLOSURE

Snap the front and back enclosure halves together. The four corner clips should click into place. Do not force it!

STEP 8 — FIRST POWER ON

Connect your GyanBit to a laptop via USB. The LED on the Pico should light up. If pre-flashed, the boot screen will appear immediately!

CONGRATULATIONS! YOU JUST BUILT A COMPUTER.

CHAPTER 6 — SETTING UP GYANBIT STUDIO

GyanBit Studio is your game development environment. It runs entirely in your web browser.

Requirements:

- Google Chrome or Microsoft Edge (version 89+)
- Laptop or Desktop (Win/Mac/Linux)
- Micro-USB Cable

Steps:

1. Go to **gyanbit.vercel.app**
2. Connect your GyanBit device via USB
3. Click **CONNECT** in the toolbar
4. Select the 'RP2040' or 'Serial Port' from the picker
5. Click **FLASH** (first time only) to install the runtime

> ***BYTE says:*** *"Firefox and Safari don't support WebSerial — the technology that lets the browser talk to USB devices. Stick to Chrome or Edge!"* 🌐

CHAPTER 7 — UNDERSTANDING THE 128X64 SCREEN

Your GyanBit screen is 128 pixels wide and 64 pixels tall. Every pixel is either ON (bright) or OFF (dark).

```
(0,0)──────────────────────────────(127,0)
  │                                   │
  │   X goes from 0 (left) to 127     │
  │   Y goes from 0 (top) to 63       │
  │                                   │
  │   bit.dot(64, 32) = center        │
  │                                   │
(0,63)─────────────────────────────(127,63)
```

Position	X	Y
Top-left	0	0
Top-right	127	0
Bottom-left	0	63
Bottom-right	127	63
Center	64	32

BYTE says: *"Use the grid paper at the back of this manual to sketch your game layouts. Real developers do this — it's called a 'level design document'!"* 📐

CHAPTER 8 — YOUR FIRST GAME

Type this into the GyanBit Studio editor:

```
let x = 64;
let y = 32;

bit.onPress('left',  () => { x = x - 3; });
bit.onPress('right', () => { x = x + 3; });
bit.onPress('up',    () => { y = y - 3; });
bit.onPress('down',  () => { y = y + 3; });

bit.loop(() => {
  bit.clear();
  bit.fill(x, y, 6, 6);
});
```

What each line does:

- `let x = 64` — Start in the middle
- `bit.onPress(...)` — Handle button input
- `bit.loop(...)` — Runs 30 times per second
- `bit.clear()` — Erase previous frame
- `bit.fill(x, y, 6, 6)` — Draw a 6x6 square

CHAPTER 9 — THE BIT.* API REFERENCE

DRAWING COMMANDS

Command	What it does
`bit.clear()`	Erase everything
`bit.dot(x,y)`	Light up one pixel
`bit.line(x1,y1,x2,y2)`	Draw a line
`bit.box(x,y,w,h)`	Draw rectangle
`bit.fill(x,y,w,h)`	Filled rectangle
`bit.circle(cx,cy,r)`	Draw circle
`bit.text(x,y,str)`	Write text

INPUT COMMANDS

Command	What it does
`bit.onPress(btn, fn)`	Call fn on press
`bit.onHold(btn, fn)`	Call fn while held
`bit.isHeld(btn)`	Is button held?

CHAPTER 10 — DEEP DIVE: GYANBIT STUDIO

GyanBit Studio supports two modes:

Block Mode (Beginners):

Visual drag-and-drop blocks. Each block represents one line of code. Perfect for starting your journey!

Text Mode (Intermediate/Advanced):

Write JavaScript directly. This is what professional developers use.

COMPOSING CHIP-TUNE MUSIC

Your speaker plays square waves. Use frequencies (Hz) to play notes:

```
// Simple C Major arpeggio
bit.melody([[523, 100], [659, 100], [784, 200]]);
```

> ***BYTE says:*** *"Middle C is 262Hz. High C is 523Hz. Double the frequency = One octave higher!"* 🎵

CHAPTER 11 — BUILT-IN GAME LIBRARY

GyanBit comes pre-loaded with 5 games. Load any of them in the Studio to study how they work!

"Don't just play these games — read the code! Press LOAD to open any game in the editor. That's how programmers learn." 📚

- 🐍 **Snake:** Classic apple-eating game. Teaches arrays and collision detection.
- 🧱 **Breakout:** Bounce a ball to break bricks. Teaches velocity and physics.
- 🪨 **Dodger:** Avoid falling rocks. Teaches random numbers and difficulty scaling.
- 🌀 **Maze:** Randomly generated mazes. Teaches recursion and grids.
- 🏓 **Pong:** Player vs AI. Teaches state management and AI behavior.

CHAPTER 12 — VARIABLES, LOOPS & LOGIC

WHAT IS A VARIABLE?

A variable is like a labeled box that holds a value.

```
let score = 0;       // A box called "score"
score = score + 1;   // Add 1 to the box
```

WHAT IS A LOOP?

A loop is something that repeats. `bit.loop()` runs 30 times a second!

WHAT IS AN IF STATEMENT?

An if statement makes a decision. "IF this is true, do this."

```
if (score > 10) {
  bit.text(30,30, "YOU WIN!");
}
```

CHAPTER 13 — GAME DESIGN 101

The Golden Rule: Start easy, get harder gradually. This is called a **difficulty curve**.

DEBUGGING LIKE A DETECTIVE

1. **Read the error:** It tells you the line number!
2. **Add `bit.print()`:** Check your variables.
3. **Comment out code:** Use `//` to hide broken parts.
4. **Simplify:** Test only the part that is broken.

> *"Bugs aren't failures — they're puzzles! Every pro programmer breaks their code constantly. Just keep solving the puzzle!"* 🕵️

CHAPTER 14 — THE HACKER'S HANDBOOK

SECRET BUTTON COMBOS

Combo	Effect
A + B (3s)	Invert screen colors
B + Up (startup)	Brightness mode
A + Down + Up	Show system info

GPIO EXPANSION

You can connect extra LEDs, sensors, or motors to the Pico's unused pins (GP0, GP8-14, etc.). Be careful and use resistors!

CHAPTER 15 — CARING FOR YOUR CONSOLE

- Keep away from water and moisture.
- Store in the box when not in use.
- Clean screen with a dry cloth only.
- Do not drop it — OLED glass is fragile!

ECO-FRIENDLY PROMISE

Our box is 100% recycled board. The electronics use lead-free solder. When you're done with GyanBit, reuse the Pico for another project!

> *"The tray inside your box can be turned into a console stand! Just fold along the lines. Reduce, Reuse, Record high scores!"* ♻️

CHAPTER 16 — TECHNICAL REFERENCE

Parameter	Value
Microcontroller	RP2040 Dual-Core ARM Cortex-M0+
RAM / Flash	264KB SRAM / 2MB Flash
Display	1.3" OLED (SH1106), 128x64
Audio	3W Class-D Amp, PWM Input
Voltage	3.3V Logic, 5V USB Input
Connectivity	Micro-USB (Serial @ 115200)

FIRMWARE UPDATES

1. Download the latest runtime from **gyanbit.vercel.app/update**.

2. Open GyanBit Studio, click OPEN and select the file.

3. Click FLASH. Done in 10 seconds!

CHAPTER 17 — TROUBLESHOOTING

Problem	Fix
Blank Screen	Check OLED is snapped in tight.
No CONNECT button	Use Chrome or Edge browser.
Device not found	Check USB cable is for data.
No sound	Check Amp and Speaker wires.
Code won't run	Click FLASH before UPLOAD.

Still stuck? Email tc26gf005@gyanbit.vercel.app!

CHAPTER 18 — ACHIEVEMENT CHECKLIST

Beginner — Circuit Tier

[ ] Assembled GyanBit from scratch
[ ] Powered on the device
[ ] Connected to GyanBit Studio

Intermediate — Code Tier

[ ] Wrote your first game loop
[ ] Drew a custom sprite
[ ] Used `bit.beep()` for sound

Advanced — Engineer Tier

[ ] Built a game from scratch
[ ] Shared a game with a friend
[ ] Used `bit.print()` to debug

CHAPTER 19 — GRID PAPER

Design your games and sprites here!

```
     0   4   8   12  16  20  24  28  32  36  40  44  48  52  56  60
0   +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
4   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
8   +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
12  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
16  +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
20  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
24  +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
28  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
32  +---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
```

8x8 Sprite Grids:

```
+--+--+--+--+--+--+--+--+
| | | | | | | | | |
+--+--+--+--+--+--+--+--+
| | | | | | | | | |
+--+--+--+--+--+--+--+--+
| | | | | | | | | |
+--+--+--+--+--+--+--+--+
| | | | | | | | | |
+--+--+--+--+--+--+--+--+
```

```
+--+--+--+--+--+--+--+--+
| | | | | | | | | |
+--+--+--+--+--+--+--+--+
| | | | | | | | | |
+--+--+--+--+--+--+--+--+
| | | | | | | | | |
+--+--+--+--+--+--+--+--+
| | | | | | | | | |
+--+--+--+--+--+--+--+--+
```

CHAPTER 20 — QUICK REFERENCE CARD

GYANBIT CHEAT SHEET

SCREEN: 128 x 64 px | Center: (64, 32)

BUTTONS: 'up', 'down', 'left', 'right', 'a', 'b'

TOP COMMANDS:

```
bit.clear()          // Erase
bit.fill(x,y,w,h)    // Draw
bit.text(x,y,str)    // Write
bit.loop(fn)         // 30 FPS
```

NOTES: C=262, D=294, E=330, F=349, G=392

HACKS: A+B (3s) = Invert Color

GyanBit v1.0 | 2026

Build It. Code It. Play It.